inTelix Solutions
the future is bright

FOR: *Domains, Web Hosting, SSL Certificates, Website Design, Strategy, Branding & Re-Branding, Photo & Videography Services*
Postal Address: *P.O Box 17723 – 00500, Nairobi, Kenya*
Information | Sales | Support (Call/ Text): *+254 710 60 42 42*
E-Mail: *info@intelixsolutions.net* | ***Website:*** *www.intelixsolutions.net*

FUNCTIONAL SPECIFICATIONS DOCUMENT (FSD):

AMAZING SOULS ORGANIC MEAL & JUICE ONLINE ORDERING PLATFORM

Document Version: 1.1

Date:

Status:

1.0 INTRODUCTION AND PROJECT VISION

This Functional Specifications Document (FSD) serves as the definitive technical and operational blueprint for the design, development, and deployment of the Amazing Souls ecommerce web application. This document outlines the project's purpose, scope, functional and non-functional requirements, and the underlying business logic that will govern its operation. It is intended to be the single source of truth for all project stakeholders, including the developer, project manager and business owner, ensuring a shared understanding of the project's objectives and deliverables.1

1.1 Project Purpose

The Amazing Souls platform is conceived as a premium B2C e-commerce web application. Its primary purpose is to connect health-conscious consumers within specific counties in Pennsylvania, USA, to a curated selection of high-quality, organic meals and freshly pressed juices. The platform will facilitate both on-demand and scheduled delivery, offering a seamless and sophisticated user experience from browsing to consumption.

1.2 Problem Statement

The project directly addresses a well-defined market need among busy professionals, fitness enthusiasts, and health-focused individuals who seek convenient access to nutritious, ready-to-eat meals. The core problem it solves is the significant time and effort investment required for meal planning, grocery shopping, and cooking. This investment often leads consumers to compromise on their health goals by relying on less nutritious fast-food options or experiencing decision fatigue when faced with meal choices. Amazing Souls aims to eliminate this trade-off between convenience and health by providing a trusted, high-quality alternative.

The strategic positioning of Amazing Souls as a "premium" service is a foundational principle that informs every aspect of this specification. This is not merely a food delivery utility; it is a lifestyle platform designed to support the health and wellness goals of its users. This premium focus differentiates the brand from mass-market food aggregators and justifies a higher price point. This decision has significant downstream implications, mandating superior non-functional requirements such as high performance and security, and driving specific functional requirements like comprehensive nutritional data, advanced dietary filtering, and an aesthetically pleasing, image-

rich user interface. The platform's success hinges on delivering an experience that feels exclusive, reliable, and aligned with the values of a discerning customer base.

1.3 Business Goals & Objectives

The overarching business goal is to establish Amazing Souls as the preeminent online platform for premium, healthy meal delivery in its target launch counties within a 24-month period. This goal is supported by a set of specific, measurable, achievable, relevant, and time-bound (SMART) objectives.[10]

- **Objective 1: User Acquisition & Retention:** Achieve a base of 500 active weekly subscribers within the first 12 months of operation.
- **Objective 2: Revenue Growth:** Attain and maintain an Average Order Value (AOV) of at least USD 35.00.
- **Objective 3: Customer Satisfaction:** Maintain a Customer Satisfaction (CSAT) score of 90% or higher, to be measured through systematic post-order feedback surveys.
- **Objective 4: Customer Loyalty:** Achieve a repeat customer rate of 40% within the first year, indicating strong product-market fit and customer loyalty.

1.4 Success Criteria

The success of the Amazing Souls project will be evaluated against a clear set of criteria that encompass both business outcomes and technical execution.[1]

- **Business Success:** The primary measure of success will be the achievement of the SMART objectives outlined in Section 1.3. This includes meeting targets for user acquisition, AOV, CSAT, and customer retention.
- **Project Success:** The project will be considered successful if it is delivered in adherence to the agreed-upon timeline and budget as defined in the project constraints (Section 8.2).
- **Technical Success:** The final deliverable must be a stable, secure, and performant platform that meets all functional and non-functional requirements detailed within this document, particularly those related to performance (Section 9.1) and security (Section 9.2).

2.0 STAKEHOLDERS AND RESPONSIBILITIES

Clear identification of project stakeholders and their respective responsibilities is critical for ensuring accountability, streamlined communication, and effective decision-making throughout the project lifecycle. The following individuals and roles constitute the core project team.

- **Project Sponsor/ Owner:** – Holds final authority on the project's budget, scope, and ultimate business goals. Responsible for championing the project within the organization and providing high-level direction. Serves as the primary business representative to the development team. Responsible for defining features, managing and prioritizing the product backlog, and providing final sign-off on the FSD and feature completion. This role is the key point of contact for all requirement clarifications.
- **Project Manager:** – Responsible for the day-to-day management of the project, including timeline adherence, resource allocation, and risk management. Facilitates communication between all stakeholders and ensures the project remains on track.
- **Lead Developer / Architect:** – Responsible for the overall technical vision and architecture of the application. Makes key decisions regarding the technology stack, database design, and development best practices. Implements and oversees the development work.

3.0 USER ROLES AND PERSONAS

To ensure a user-centric design process, the system defines distinct user roles with specific permissions. These roles are brought to life through detailed user personas, which represent the target audience and guide feature development by focusing on real-world goals and frustrations.[11]

3.1 User Roles & Permissions Matrix

The system will support four primary user roles, each with a distinct set of permissions that govern their access to and interaction with the platform's functionalities.[1]

Role	Description	Key Permissions
Guest	An unauthenticated visitor to the web application.	Enter a zip code to verify service availability; Browse all products and categories (if zip code is valid); View informational pages (e.g., About Us, FAQ); Utilize search and filtering functionalities; Cannot add items to a cart, initiate checkout, or save personal information.
Customer	An authenticated user who has registered an account.	All Guest permissions; Create and manage a user profile; Save multiple delivery addresses and payment methods; Add items to cart and complete checkout; View comprehensive order history; Submit product ratings and reviews; Manage communication preferences.
Admin	A superuser with full backend access, typically an employee of Amazing Souls.	Full control over all aspects of the platform; Manage user accounts (Customers and Partners); Full CRUD (Create, Read, Update, Delete) capabilities for products, categories, and menus; Manage all orders; Create and manage promotional campaigns; Define and adjust delivery zones and fees by zip code; View and export all system reports and analytics; Configure system settings.
Partner	A user from a partner kitchen or restaurant with limited backend access.	View and manage menu items assigned to their specific location; View and process incoming orders routed to their kitchen; Update the status of orders (e.g., "Confirmed," "Preparing," "Ready for Pickup"); Cannot access global settings, financial reports, or customer data outside of their assigned orders.

3.2 User Personas

The following personas are fictional representations of the primary user types, based on market research, to guide design and development decisions.[16]

3.2.1 Persona 1: Alex, the Busy Professional

- **Demographics:** 32-year-old male, working as a Marketing Manager in Philadelphia, PA. He is tech-savvy, has a high disposable income, but is extremely time-poor due to a demanding career.
- **Goals:** Alex's primary goal is to consume quick, healthy, and high-quality meals without the time commitment of cooking or the "decision fatigue" that comes with browsing endless, generic food delivery options.[6] He values efficiency, predictability, and premium quality. He wants a "set it and forget it" solution for his daily nutrition.
- **Frustrations:** He is tired of the unhealthy, greasy options that dominate the conventional takeout market. He has tried meal prep kits but finds they still require too much time and effort. He is frustrated by slow websites, complicated checkout processes, and unreliable delivery times.
- Key Needs from Amazing Souls:
  - An exceptionally fast and intuitive ordering process that minimizes clicks and decisions.[18]
  - The ability to create subscription or recurring order plans for his work week.
  - Accurate, real-time order tracking with reliable Estimated Times of Arrival (ETAs) to fit into his tight schedule.[19]
  - Functionality to save "favorite" meals and reorder them with a single click.

3.2.2 Persona 2: Maya, The Health-Conscious Planner

- **Demographics:** 28-year-old female, working as a fitness instructor and nutritionist in Philadelphia, PA. She is extremely health-conscious, follows a specific diet (e.g., Ketogenic), and is a meticulous planner regarding her nutritional intake.
- **Goals:** Maya's goal is to strictly adhere to her diet and fitness regimen by consuming meals with precise nutritional profiles. She demands complete transparency about ingredients and macronutrient content to make informed choices that align with her health objectives.
- **Frustrations:** She finds it nearly impossible to get accurate and detailed nutritional information from most restaurants and delivery services. "Healthy" labeled options are often bland, misleading, or contain hidden ingredients (like sugars or unhealthy oils) that disrupt her diet.[3]
- Key Needs from Amazing Souls:
  - Advanced and multi-select filtering capabilities for dietary plans (e.g., Vegan, Keto, Paleo), allergens (e.g., Gluten-free, Dairy-free, Nut-free), and specific nutritional values (e.g., calories, protein, carbohydrates).[7]
  - Complete, accurate, and easily accessible ingredient lists and macronutrient breakdowns (protein, fat, carbs) for every single item on the menu.[21]
  - High-quality, realistic photographs that accurately represent the meal's portion size and composition.[8]
  - The ability to customize orders with clear modifiers (e.g., "no dressing," "extra protein," "sauce on the side").

3.2.3 Persona 3: Lucas, The Platform Administrator

- **Demographics:** 40-year-old Operations Manager for the Amazing Souls business. Responsible for the smooth execution of all daily activities.
- **Goals:** Lucas's primary goal is to ensure seamless day-to-day operations. This includes managing inventory efficiently to prevent waste, monitoring business performance to make data-driven decisions, resolving customer issues quickly, and ensuring orders are fulfilled accurately and on time.
- **Frustrations:** Lucas is frustrated by the inefficiency and error-prone nature of juggling multiple, disconnected systems for order management, menu updates, and customer communications.[22] He lacks a single, real-time view of the business's health, and manual data entry frequently leads to costly mistakes.
- Key Needs from Amazing Souls:
  - A centralized administrative dashboard that displays Key Performance Indicators (KPIs) at a glance, such as total sales, order volume, new vs. repeat customers, and AOV.[23]
  - An efficient and integrated Order Management System (OMS) to track every order from the moment it is placed until it is successfully delivered.[25]
  - An intuitive Menu Management System (MMS) that allows for instant updates to items, pricing, and availability, with changes reflecting immediately on the live website.[21]
  - Robust reporting and analytics tools to generate insights on sales trends, product performance, and customer behavior, enabling strategic business planning.[27]

4.0 USE CASES AND USER SCENARIOS

This section outlines key scenarios illustrating how the defined user personas will interact with the Amazing Souls platform to achieve their specific goals. These use cases serve as practical narratives that inform the detailed functional requirements in the subsequent section.[11]

- UC-00: Pre-Shopping Delivery Area Validation
  - **Actor:** A new visitor (Guest)
  - **Scenario:** A visitor lands on the Amazing Souls homepage. A prominent notification prompts them to enter their zip code to check for delivery availability. The user enters their zip code. If it's within a serviceable county, they are granted access to browse the menu. If not, they receive a polite message indicating that service is not yet available in their area, with an option to sign up for future notifications.
- UC-01: First-Time Order with Guest Checkout
  - **Actor:** Alex, The Busy Professional
  - **Scenario:** Alex lands on the Amazing Souls homepage during his lunch break and validates his zip code. He quickly browses the "Lunch" category, selects a "Spicy Chicken & Quinoa Bowl," proceeds directly to checkout, enters his office address and credit card details, and places the order for immediate delivery—all without being forced to create an account.
- UC-02: Creating a Weekly Subscription Plan
  - **Actor:** Maya, The Health-Conscious Planner
  - **Scenario:** Maya logs into her account. She navigates to the "Meal Plans" section, filters for "Keto" and "Gluten-Free," and selects a plan for 5 lunches and 5 low-sugar juices per week. She customizes her delivery days (Monday-Friday) and time slot (12 PM - 1 PM), saves her credit card for recurring payments, and confirms her subscription.
- UC-03: Re-ordering a Favorite Meal
  - **Actor:** Alex, The Busy Professional
  - **Scenario:** Alex logs into his account and navigates to his profile dashboard. He sees a "Favorites" or "Recent Orders" section, locates the "Grilled Salmon with Asparagus" he enjoyed last week, and clicks a "Re-order" button. The system takes him directly to the checkout page with the item already in his cart and his default address and payment method pre-selected.
- UC-04: Searching and Filtering for a Specific Dietary Need
  - **Actor:** Maya, The Health-Conscious Planner
  - **Scenario:** Maya uses the search bar to look for "vegan." On the results page, she applies filters from the sidebar: she ticks the "Gluten-Free" checkbox and uses a slider to set the "Protein" range to "> 20g." The product list updates in real-time to show only the meals that meet all three of her criteria.
- UC-05: Processing a New Order
  - **Actor:** Lucas, The Platform Administrator
  - **Scenario:** A new order alert sounds on the admin dashboard. Lucas clicks on the notification and views the order details: customer name, delivery address, items ordered, and payment confirmation. He clicks "Accept Order," and the system automatically prints a ticket in the kitchen and updates the order status to "Being Prepared" on the customer's end.
- UC-06: Updating a Menu Item's Availability
  - **Actor:** Lucas, The Platform Administrator
  - **Scenario:** The head chef informs Lucas that they have run out of fresh avocados. Lucas logs into the

admin portal, navigates to the Menu Management System, finds the "Avocado Power Bowl," and changes its status from "In Stock" to "Out of Stock." The item immediately appears as unavailable on the live customer-facing website, preventing any further orders for it.

5.0 FUNCTIONAL REQUIREMENTS

This section forms the core of the FSD, providing a detailed, feature-by-feature breakdown of what the Amazing Souls system must be capable of doing. Requirements are presented in a user story format ("As a [user type], I want to [action] so that [benefit]") to maintain a clear focus on user value.[2]

5.1 Geographic & Delivery Zone Features

This module covers all functionalities related to service area validation, which is critical for managing customer expectations and operational boundaries.

- FR-GEO-01: Zip Code-Based Service Validation
  - As a guest user, I want to enter my zip code upon visiting the site to immediately confirm if delivery is available in my area, so I don't waste time browsing products I can't order. The system must display a clear, non-intrusive notification or modal for this purpose. [60]
- FR-GEO-02: Address Autocomplete Scoping
  - As a user entering my delivery address, I want the Google Maps address autocomplete suggestions to be restricted to only show addresses within the pre-approved service areas, so that I am guided to select a valid, deliverable address.

5.2 User Account & Profile Management

This module covers all functionalities related to user identity, personalization, and data management, which are essential for a tailored and efficient user experience.[6]

- FR-UAM-01: User Registration
  - As a new user, I want to create an account using my email address and a secure password, or by linking my existing Google or Facebook account, so that I can have a personalized experience and save my information for future orders. The system must validate email format and enforce password complexity rules.
- FR-UAM-02: Secure Login
  - As a returning user, I want to log in securely with my credentials or social media account to access my profile, view my order history, and place new orders quickly. The system must protect against brute-force login attempts through mechanisms like rate limiting.
- FR-UAM-03: Password Management
  - As a user who has forgotten my password, I want to request a password reset via a secure, time-sensitive link sent to my registered email address, so that I can regain access to my account without compromising its security.
- FR-UAM-04: Profile Management
  - As a logged-in user, I want to access a personal dashboard where I can view and edit my core profile details, including my first name, last name, and primary phone number.
- FR-UAM-05: Address Management
  - As a logged-in user, I want to save, edit, and delete multiple delivery addresses, each with a custom label (e.g., "Home," "Office"), so that I can select my desired delivery location quickly during checkout

without re-typing information.[30]

- FR-UAM-06: Payment Management
  - As a logged-in user, I want to securely save, view (with masked details, e.g., "Visa ending in 1234"), and remove my credit/debit card information, so that I can complete future payments with a single click. All payment data handling must be PCI DSS compliant.
- FR-UAM-07: Order History
  - As a logged-in user, I want to view a comprehensive list of my past and current orders, sortable by date and status. Each entry should display the order number, date, total cost, items purchased, and current status (e.g., "Delivered," "In Transit"), allowing me to track my spending and easily re-order past purchases.[29]
- FR-UAM-08: Communication Preferences
  - As a user, I want a settings page where I can manage my communication preferences, allowing me to opt in or out of different types of notifications (e.g., promotional emails, order update SMS, marketing push notifications) independently.

5.3 Product Catalog & Menu Presentation

The presentation of the menu is a critical component of the premium user experience. It must be visually appealing, informative, and easy to navigate.[9]

- FR-PCM-01: Dynamic & Categorized Menu
  - As a user, I want to browse menu items that are logically organized into clear, intuitive categories (e.g., "Breakfast," "Lunch," "Juices," "Meal Plans," "Weekly Specials") so that I can easily navigate the offerings and find what I am looking for without feeling overwhelmed.[8]
- FR-PCM-02: High-Quality Visuals
  - As a user, I want to see professional, high-resolution, and appetizing photos for every menu item. The use of short, well-produced videos showcasing the meal's preparation or presentation is also required for featured items. Visuals must be realistic and accurately represent the product to build trust and entice orders.[8] PDF menus are strictly forbidden as they provide a poor mobile experience.[9]
- FR-PCM-03: Detailed Product Information Page
  - As a user, I want to click on any menu item to view a dedicated product page. This page must contain a compelling and descriptive narrative, a complete and unambiguous list of all ingredients, a detailed nutritional information table (including calories, protein, fat, carbohydrates per serving), and prominent, clear allergen warnings (e.g., "Contains: Nuts, Dairy, Gluten").[7]
- FR-PCM-04: Customer Ratings and Reviews
  - As a user, I want to see an average star rating and read reviews from other verified purchasers directly on the product page. This social proof is crucial for building trust and helping me make an informed decision, especially when trying a new item.[29]
- FR-PCM-05: Real-Time Item Availability
  - As a user, I want to clearly see the availability status of each item (e.g., "In Stock" or a visually distinct "Sold Out" label) directly on the menu and product pages. This prevents the frustration of adding an item to my cart only to find out it's unavailable at checkout.

5.4 Search, Filtering, & Discovery

Effective search and filtering tools are essential for users like Maya to find products that meet their specific needs, and for users like Alex to reduce decision fatigue.[6]

- FR-SFD-01: Intelligent Keyword Search
  - As a user, I want a prominent search bar, accessible from all main pages, that provides real-time, relevant results and autocomplete suggestions as I type. The search algorithm should be fault-tolerant, handling common misspellings and recognizing keywords related to ingredients and dish names.
- FR-SFD-02: Advanced Multi-Select Filtering
  - As a user, I want a persistent filtering panel (e.g., a left sidebar on desktop) that allows me to refine the product list by selecting multiple criteria simultaneously. This must include filters for:
    - **Dietary Type:** Vegan, Vegetarian, Keto, Paleo.
    - **Allergens:** Gluten-Free, Dairy-Free, Nut-Free.
    - **Nutritional Range:** Sliders or input fields for Calories, Protein, and Carbohydrates.
    - **Price Range:** A slider to filter items within a specific price bracket.
- FR-SFD-03: Personalized Recommendations
  - As a logged-in user, I want to see a "Recommended for You" or "You Might Also Like" section on the homepage and product pages. Initially, this will be rule-based (e.g., showing other keto items to a user who frequently orders keto meals), with the architecture allowing for future machine learning enhancements.

5.5 Shopping Cart & Ordering Process

The transition from browsing to buying must be as seamless and frictionless as possible to minimize cart abandonment.

- FR-SCO-01: Add to Cart Functionality
  - As a user, I want to easily add items to my shopping cart from both the main menu grid and the individual product detail page with a single click. The cart icon should provide immediate visual feedback, such as updating a badge with the number of items.
- FR-SCO-02: Cart Management
  - As a user, I want to be able to view my shopping cart at any time. Within the cart view, I must be able to easily adjust the quantity of each item, remove items completely, and see the subtotal update in real-time.
- FR-SCO-03: Order Customization and Special Instructions
  - As a user, I want the ability to customize my order. For applicable items, this will be through predefined modifiers (e.g., Protein Choice: Chicken/Tofu). Additionally, a "Special Instructions" text field must be available for each item to accommodate specific requests like "Dressing on the side" or "No onions".[29]
- FR-SCO-04: Guest Checkout
  - As a new or infrequent user, I want a prominent and clearly labeled "Guest Checkout" option. This is critical to reduce friction and prevent abandonment for users who are not ready to commit to creating an account. After a successful guest order, the user should be offered a simple option to create an account using the details they just provided.
- FR-SCO-05: Delivery Scheduling
  - As a user, I want to select my preferred delivery date and time during checkout. The system must present available delivery slots (e.g., "Today, 1:00 PM - 2:00 PM," "Tomorrow, 6:00 PM - 7:00 PM") and

prevent selection of unavailable or past slots.

5.6 Payment & Checkout

The payment process must be secure, transparent, and accommodate local payment habits to maximize conversion rates.

- FR-PAY-01: Secure and Localized Payment Gateway Integration
  - The system shall integrate with multiple secure, PCI DSS compliant payment gateways suitable for the US market. This must include leading providers such as **Stripe**, **PayPal**, and **Square** to ensure broad card and digital wallet compatibility.
- FR-PAY-02: Multiple Payment Methods
  - As a user, I want the flexibility to pay for my order using a variety of methods. The system must support:
    - Credit/Debit Cards (Visa, Mastercard).
    - Digital Wallets supported by the gateways (e.g., Apple Pay, Google Pay).
- FR-PAY-03: Promotions and Loyalty Points Application
  - As a user, I want a clear input field in the checkout process to apply a valid promotional code. Upon application, the discount must be immediately reflected in the order summary. Similarly, if a loyalty program is active, I should have the option to redeem my points for a discount.
- FR-PAY-04: Transparent Cost Summary
  - As a user, before I commit to payment, I want to see a clear, unambiguous, and itemized breakdown of all costs on the final checkout page. This summary must include: a subtotal of all items, applicable taxes (e.g., VAT), the calculated delivery fee for my address, any discounts applied, and the final, bolded total amount to be charged.[14]

5.7 Order Management & Fulfillment (Customer-Facing)

Post-purchase communication and transparency are key to building customer trust and reducing support inquiries.

- FR-OMF-01: Multi-Channel Order Confirmation
  - Upon successful payment authorization, I want to receive an immediate order confirmation. This confirmation must be displayed on-screen, sent to my registered email address with a full order summary and receipt, and optionally sent as a summary via SMS if I have opted in.
- FR-OMF-02: Real-Time Order Status Updates
  - As a user, I want to be able to log into my account and view the real-time status of my active order in my dashboard. The status should progress through clear stages: "Order Confirmed," "Being Prepared," "Out for Delivery," and "Delivered".
- FR-OMF-03: Proactive Push Notifications
  - As a user with the (future) mobile app or browser notifications enabled, I want to receive proactive push notifications for key order milestones. The most critical notifications are "Your order is out for delivery" and "Your driver is arriving soon".
- FR-OMF-04: Live GPS Driver Tracking
  - Once my order status changes to "Out for Delivery," I want to access a feature that displays the

delivery driver's current location in real-time on an embedded map. This map must also show my delivery location and provide a dynamically updating Estimated Time of Arrival (ETA) based on the driver's progress and traffic data.

5.8 Admin & Partner Backend

The backend portal is the central nervous system of the operation, enabling efficient management of all business functions for administrators like Lucas.

- FR-ADM-01: Centralized Performance Dashboard
  - As an Admin (Lucas), upon logging in, I want to see a centralized dashboard that provides a real-time, at-a-glance overview of Key Performance Indicators (KPIs). This must include widgets for: Total Sales (today/week/month), Number of Orders, Average Order Value (AOV), New vs. Repeat Customers, and a list of the top-selling products.
- FR-ADM-02: Integrated Order Management System (OMS)
  - As an Admin or Partner, I want a comprehensive OMS where I can view a list of all incoming and active orders. The interface must allow filtering by status (New, Confirmed, In-Progress, etc.) and provide clear visual and audible alerts for new, unconfirmed orders. I must be able to accept or reject new orders, with the system automatically triggering the next step in the workflow (e.g., notifying the customer, sending the order to the kitchen display).
- FR-ADM-03: Integrated Menu Management System (MMS)
  - As an Admin, I want a user-friendly MMS to perform full CRUD operations on all menu items and categories. This system must allow me to update names, descriptions, pricing, high-quality imagery, detailed nutritional data, and, crucially, the inventory status (e.g., toggling an item between "In Stock" and "Sold Out"). Changes made in the MMS must reflect on the live customer-facing website instantly.
- FR-ADM-04: Customer Relationship Management (CRM)
  - As an Admin, I want to access a customer management module where I can look up customer profiles by name, email, or phone number. Viewing a customer's profile should show their contact information and complete order history, enabling me to provide informed and efficient customer support.
- FR-ADM-05: Promotions Management
  - As an Admin, I want a module to create and manage promotional campaigns. I must be able to generate discount codes (e.g., "PHILLY20" for 20% off) and define their parameters, such as percentage or fixed amount, minimum order value, valid-from and expiry dates, and total usage limits.
- FR-ADM-06: Delivery Zone Management
  - As an Admin, I want a management interface to define our service area by adding, editing, and deleting a list of supported zip codes for specific counties in Pennsylvania. For each distinct zip code or group of zip codes, I must be able to set a specific delivery fee.
- FR-ADM-07: Reporting & Analytics
  - As an Admin, I want a powerful reporting module to generate and export detailed reports in CSV format. This must include reports on:
    - **Sales:** By product, category, time period (day, week, month), and sales channel.
    - **Customers:** New vs. repeat customer analysis, customer lifetime value (CLV).
    - **Inventory:** Sell-through rates and identification of slow-moving items.

A critical element of the backend design is the deep, real-time integration between the Order Management

System (OMS), Menu Management System (MMS), and a new, implied module for ingredient-level Inventory Management.

A superficial implementation would treat these as separate silos. However, for the operational efficiency required by a premium service, these systems must form an automated feedback loop.

When a customer places an order containing a "Grilled Salmon" meal via the OMS, the system must do more than just record the sale.

It must immediately access the recipe data associated with that meal and decrement the corresponding raw ingredients from the central inventory database (e.g., -1 salmon fillet, -100g quinoa, -50g broccoli). This action, in turn, must trigger a check against inventory thresholds.

If the number of available salmon fillets drops to zero, the inventory system must signal the MMS to automatically update the status of the "Grilled Salmon" menu item to "Out of Stock" on the live website. This automated, interconnected workflow is non-negotiable. It prevents the operational failure of selling unavailable products, eliminates the need for manual inventory checks by the admin (Lucas), and enhances the customer experience by preventing post-order cancellations due to stock issues.

6.0 BUSINESS RULES

This section defines the explicit operational logic, constraints, and policies that the system must enforce to ensure consistency, prevent errors, and align with business strategy.

- **BR-01: Geographic Ordering Constraint:** The system will prohibit the placement of any order for a delivery address with a zip code that is not on the admin-managed list of supported zip codes. Users attempting to do so will receive a clear error message.
- **BR-02: Price Calculation Formula:** The final, chargeable price for any order must be calculated according to the strict formula: Final Price = (Sum of Item Prices) + (Applicable VAT) + (Delivery Fee for the specific zone) - (Value of Applied Discount). All components of this calculation must be displayed transparently to the user at checkout.
- **BR-03: Promotion Exclusivity:** A user may apply only one promotional code per order. The system will disallow the application of a second code if one is already active. An exception can be made only if a specific promotion is explicitly marked as "stackable" by an Administrator in the promotions management module.
- **BR-04: Inventory-Based Order Confirmation:** The system will prevent an order from being confirmed and sent for payment if any of the items in the customer's cart have a real-time inventory status of "Out of Stock." The user will be prompted to remove the unavailable item before proceeding.
- **BR-05: Order Scheduling Logic:** Users can schedule orders for delivery up to a maximum of seven (7) calendar days in advance. For same-day deliveries, an order must be placed at least 90 minutes before the beginning of the selected one-hour delivery time slot.
- **BR-06: Payment-Based Order Confirmation:** An order's status will only transition from "Pending" to "Confirmed" after the system receives a successful payment authorization response from the selected payment gateway. Failed payments will not result in a confirmed order.

7.0 PROJECT BOUNDARIES AND EXCLUSIONS

This section provides a clear and unambiguous definition of the project's scope for its initial release (Version 1.0). Its purpose is to manage stakeholder expectations and prevent "feature creep," which can jeopardize project timelines and budgets.

7.1 In-Scope for Version 1.0

The following features and functionalities are included in the initial launch of the Amazing Souls platform:

- The complete customer-facing responsive web application, fully functional on desktop, tablet, and mobile browsers.
- The complete backend portal for Administrators and Partners, including all management features detailed in Section 5.8.
- All functional requirements as explicitly detailed in Section 5.0 of this document.
- Integration with the following specified payment gateways: Stripe, PayPal, and Square.
- System-triggered transactional notifications via email and SMS for critical events (e.g., order confirmation, password reset).

7.2 Out-of-Scope for Version 1.0

The following features and functionalities are explicitly excluded from the Version 1.0 release and are designated for consideration in future project phases:

- **Native Mobile Applications:** Dedicated, installable applications for iOS and Android are planned for Phase 2. The initial launch will be a web-only experience.
- **Dedicated Driver Application:** Delivery personnel will initially manage assignments via a simplified web-based interface or through integration with a third-party logistics (3PL) provider. A native driver app with advanced features is not in scope for V1.0.
- **Third-Party Aggregator Integration:** The system will not integrate with external food delivery platforms like Uber Eats, DoorDash, or Grubhub in its initial release. All orders will originate from the Amazing Souls platform itself.
- **Advanced Recommendation Engine:** The initial "Recommended for You" feature will be based on simple, predefined rules. A more advanced engine utilizing machine learning for predictive personalization is a future enhancement.
- **In-App Customer Support Chat:** Live chat functionality will not be included in V1.0. Customer support will be provided through email and phone channels as listed on the website.

8.0 RISKS, ASSUMPTIONS, AND CONSTRAINTS

This section documents the known risks, foundational assumptions, and immovable constraints that will influence the project's execution. Proactively identifying these factors is crucial for effective project management and mitigation planning.

8.1 Assumptions

These are conditions believed to be true for the project to succeed, but which have not been definitively proven.

- **A-01: User Connectivity:** It is assumed that all users, including customers and backend administrators, will have access to a stable and reasonably fast internet connection sufficient for using a modern web application.
- **A-02: Third-Party API Reliability:** It is assumed that all third-party APIs (including Payment Gateways and Google Maps for delivery tracking and zone management) will be consistently available, performant, and provide comprehensive, accurate documentation to facilitate integration.
- **A-03: Content Provision:** It is assumed that all necessary content, including professional-grade product photography, compelling marketing copy, detailed descriptions, and accurate nutritional information, will be provided by the business stakeholders in a timely manner prior to the launch of related features.
- **A-04: Resource Availability:** It is assumed that all key personnel identified in the Stakeholders section (Section 2.0) will remain available and committed to the project throughout its lifecycle.

8.2 Constraints

These are real-world limitations or restrictions that the project must operate within.

- **C-01: Project Budget:** The project must be designed, developed, tested, and deployed within the strict, non-negotiable budget of _____.
- **C-02: Project Timeline:** The project must achieve the launch of Version 1.0 within a period, commencing from the date of formal sign-off on this FSD.
- **C-03: Technology Stack:** The application will be developed using a specific technology stack. The backend will be built with the **PHP** programming language using the **Laravel Framework**. Frontend interactivity will be handled by **Livewire and Alpine.js**. The database will be **MySQL**. The application will be hosted on a **Digital Ocean Droplet**, and all media files (e.g., product images, videos) will be stored using **Digital Ocean Spaces**, which provides an S3-compatible object storage solution.

8.3 Risks

These are potential events or conditions that could negatively impact the project's timeline, budget, or overall success. Each risk should have a corresponding mitigation strategy developed by the Project Manager.

- **R-02: Key Resource Dependency:** The unforeseen departure or extended unavailability of a key team member (e.g., the Lead Developer) could significantly impact the project timeline and knowledge continuity.
- **R-03: Regulatory Changes:** There is a risk that changes in Pennsylvania's food safety regulations for

Limited Food Establishments or other relevant state laws could occur during the development lifecycle, requiring unplanned scope changes to ensure legal compliance.

- **R-04: Market Adoption Rate:** There is a business risk that user adoption and order volume may be slower than projected in the initial months post-launch, impacting the achievement of revenue targets.

9.0 NON-FUNCTIONAL REQUIREMENTS (NFRS)

This section defines the quality attributes and characteristics of the system. While functional requirements define *what* the system does, NFRs define *how well* it does it. These are paramount for delivering the "premium" experience promised by the Amazing Souls brand and are not negotiable.

A core principle guiding these NFRs is the direct link between platform quality and business success. For example, robust accessibility is not merely a compliance checkbox but a strategic business decision. By adhering to WCAG 2.1 AA standards, Amazing Souls not only mitigates legal risks associated with digital discrimination lawsuits, which are increasingly common in ecommerce, but also expands its total addressable market. Furthermore, many accessibilities best practices—such as logical heading structures, high-contrast design, and keyboard navigability—directly improve the user experience and SEO for all users, reinforcing the platform's premium positioning and contributing to customer acquisition and retention.

9.1 NFR Specification Table

Category	Requirement ID	Requirement Description	Metric / Standard	Priority
Performance	NFR-PERF-01	Page Load Time	Critical user-facing pages (Homepage, Menu pages, Checkout process) must achieve a full, interactive load in under 3 seconds on a standard 4G mobile connection. Studies show that bounce rates increase by over 32% as load time goes from 1s to 3s.	Must-Have
	NFR-PERF-02	Server Response Time	The Time to First Byte (TTFB) for any server request must be under 200 milliseconds. This is a key indicator of server responsiveness and a prerequisite for fast page loads.	Must-Have
	NFR-PERF-03	Concurrency	The system must be able to handle a load of at least 200	Must-

		Handling	concurrent active users performing various actions (browsing, adding to cart, checking out) without performance degradation (i.e., response times increasing by more than 10% from the baseline).	Have
Security	NFR-SEC-01	Secure Coding Practices	All application code must be developed following secure coding standards that mitigate the risks identified in the most recent OWASP Top 10 list, including prevention of SQL Injection, Cross-Site Scripting (XSS), and insecure deserialization.	Must-Have
	NFR-SEC-02	Data Encryption	All data transmitted between the client and server (in transit) must be encrypted using Transport Layer Security (TLS) 1.2 or higher. All sensitive user data stored in the database (at rest), especially passwords, must be securely hashed using a modern, strong algorithm (e.g., bcrypt) and salted.	Must-Have
	NFR-SEC-03	Payment Security Compliance	All payment processing and handling of cardholder data must be fully compliant with the Payment Card Industry Data Security Standard (PCI DSS). No raw credit card or PAN data shall be stored on Amazing Souls application	Must-Have

			servers.	
	NFR-SEC-04	Authentication & Authorization	User authentication mechanisms must include protection against brute-force attacks, such as account lockout policies and rate limiting on login attempts. Password complexity rules (minimum length, character types) must be enforced. The principle of least privilege must be applied to all user roles.	Must-Have
Accessibility	NFR-ACC-01	WCAG Conformance	The entire customer-facing web application must conform to the Web Content Accessibility Guidelines (WCAG) 2.1 at the Level AA success criteria. This is the standard referenced in most international accessibility regulations and is the benchmark for compliance with the **Americans with Disabilities Act (ADA)** in the United States.	Must-Have
	NFR-ACC-02	Keyboard Accessibility	All interactive elements, including links, buttons, form fields, and menu items, must be fully navigable and operable using only a keyboard. A logical tab order and a visible focus indicator are mandatory.	Must-Have

	NFR-ACC-03	Screen Reader Compatibility	The website must be fully compatible with modern screen reader software (e.g., JAWS, NVDA, VoiceOver). This requires the correct semantic use of HTML, ARIA labels for complex components, and descriptive alt text for all meaningful images.	Must-Have
Scalability	NFR-SCL-01	Horizontal Scalability	The application architecture must be designed to support horizontal scaling. It should be capable of handling a projected 5x increase in user load over the first 12 months of operation by adding more server instances, without requiring a fundamental re-architecture.	Should-Have
Usability	NFR-USA-01	Mobile-First Responsive Design	The application must be built using a mobile-first, responsive design approach. It must provide an optimal and consistent user experience across all common device viewports, including mobile phones, tablets, and desktops.	Must-Have

10.0 SYSTEM CONFIGURATIONS

This section details the initial setup and configuration parameters required to make the Amazing Souls system fully operational after its deployment to a hosting environment. These are one-time or infrequent setup tasks typically performed by the development or operations team.[11]

- **CONF-01: Initial Administrator Account Creation:** A secure, command-line or one-time-use interface must be provided for creating the first super-administrator account. This process should not expose credentials in logs or plain text.
- **CONF-02: Payment Gateway Configuration:** The admin panel must provide a secure interface for entering and managing API keys, secret keys, and webhook endpoint URLs for all integrated payment gateways (Stripe, PayPal, Square). These credentials must be stored encrypted.
- **CONF-03: Third-Party Service Integration:** The system must allow for the configuration of API keys and other necessary credentials for external services, including:
  - Transactional email service (e.g., Mandrill, SendGrid) for sending order confirmations and notifications.
  - SMS gateway service for sending SMS alerts.
  - Google Maps API for geolocation and mapping features.
- **CONF-04: Initial Data Seeding:** A documented process or script must be available to populate the database with essential initial data that is required for the application to function correctly. This includes default menu categories, initial delivery zones and fees, and applicable tax rates.
- **CONF-05: Environment Variable Management:** All environment-specific configurations (e.g., database connection strings, API endpoints, secret keys, debug flags) must be managed through environment variables. A template file (e.g., env.example) must be provided in the codebase, documenting all required variables for development, staging, and production environments.

11.0 APPROVALS

This section serves as the formal sign-off for the Functional Specifications Document. The signatures below indicate that the key stakeholders have reviewed, understood, and approved the contents of this document. This approval formalizes the FSD as the definitive project blueprint and provides the official authorization to commence the design and development phases of the project.

STAKEHOLDER NAME	TITLE	SIGNATURE	DATE

-END-